VOLUMETRY & LIGHT

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Introduction

One of the most common questions I get are similar to the following: „**Where do I place the reflection?**“ and „**How do I know where to put my highlights?**“ or „**How do you know where to paint the light?**“ and in this tutorial we will learn how to do that.

WHY should I care?

Unless you are using only basecoats and edge highlights, **painting light and defining all the different shapes is majority of the painting process.** Rendering of the parts can vary (textured, smooth, glossy, reflective, matte....), but to even start, you have to know which parts of the miniature should be exposed to light and which ones should be in shade. Properly highlighted and shaded miniatures will have a better impression of 3D-likeness.

What you will learn

- One VERY easy method to find reflections
- One more complex and interesting way to find them
- How shapes and light interact

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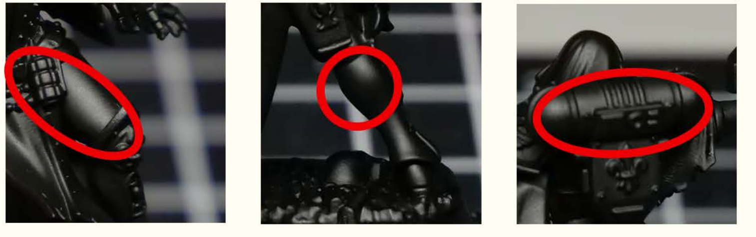
Finding reflections the easy way

The easiest way to know where to paint the reflections is to take a photo of your miniature in a black primer and copy them accordingly. Before taking the photos, make sure that your black primer is at least somewhat glossy (Chaos Black primer) and that you place your mini close under your lamp.

At the very least take a photo from the front and back, but you can take more angles to see how the light interacts with all the shapes. This has been described in multiple of my videos.

Now you can simply copy the reflections as they are formed on the surface. For example, notice the reflections on the thighs, calves or the backpack.



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Finding reflections the easy way

In the final result, you'll see that I've copied the reflection placement from the photos.






The biggest advantage of this method is that it's easy and almost always correct - at least when it comes to the reflection placement. We could easily use only this method to know where to place the light and achieve fantastic results. However, there are **3 big downsides** which we have to work around to get even better (and sometimes more correct) result.

Glossy Skull Problem

Since we are copying the reflections from a primed miniature, the reflectivness/glossiness of all parts is the same. However, when we copy the reflections in the exact same way, it can lead to errors - for example, **you wouldn't expect a skull to be as reflective as metallics**. For that reason, we have to think about the material thats being represented. In the following example, you can see that I did not paint the skull exactly as in the reference because...**skulls are typically matte, not glossy**.


Another problem with this method is that it always assumes **frontal light source** (not from above or from any other side). Lastly, without any other knowledge, we can't see where to paint the **secondary reflections.**

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Understanding volumes

If we want to play around with the light direction, multiple light sources and secondary reflections, we gotta understand shapes and how they reflect the light in general.

Above we see pictures of a **sphere** and a **cube**. The sphere reflects light in a circular pattern and depending on it's material and light conditions, we would also paint bounced light from the bottom. Furthermore, you can see that the light source is not just frontal - in fact there
, there are 2 light sources; one front top left and the other one from mid right.

The cube reflects light in a more simple way. Flat surfaces have a plain gradient that doesn't form any patterns.

The 2 other basic shapes are a **cylinder** and a **cone**. These behave very similarly when it comes to reflecting the light. In both pictures we see that the reflections are stretched from the bottom to the top, however, the cone's reflection becomes more narrow as it approaches the tip.

We have all seen how these shapes reflect the light in real life, but perhaps without thinking about it too deeply. Now with this realization fresh in mind, you can see that even the reflections found by the „Black primer-photo approach" respect the basic shapes. Notice, how the backpack reflection copies the cylinder in the middle and the two **modified spheres** on the sides.

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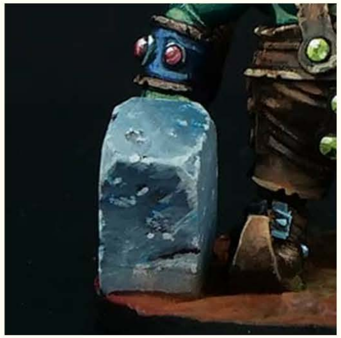
Understanding volumes

With this basic understanding, you can look at more miniatures and see how each part is painted according to it's shape. Here are some obvious examples from my minis.



Let's start with the **top left** picture. The front of the gold goblin helmet represents a slightly curved cone.

The **top right** picture shows us a pair of boobs. Although not as reflective, both are painted as recognizable spheres and the light is painted in a round pattern.

The **bottom left** represents perfect cylinders. You'll see that most gun barrels are painted in a similar fashion.

The **bottom right** picture captures a stone block. Even though it's not a perfect cube, most flat surfaces follow a similar pattern. Depending on the material and the light conditions, you have to fill them with a gradient of some kind.

Modified shapes

What shape is the body of this crocodile?

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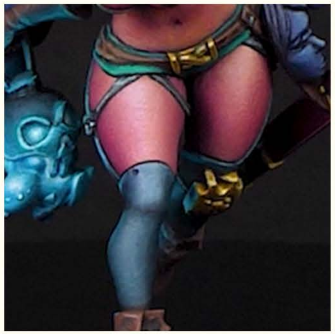
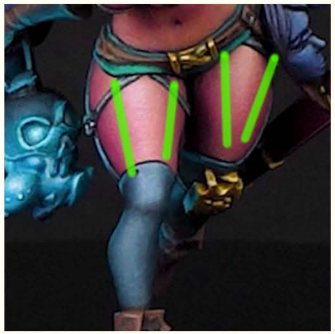
Modified shapes

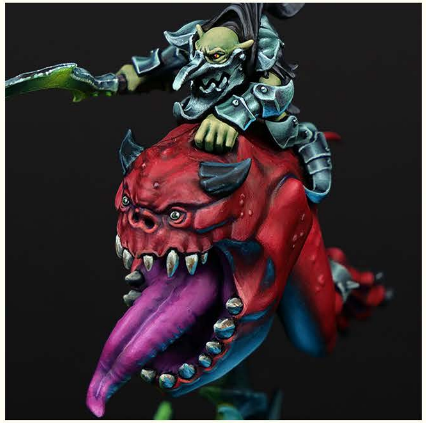
When you really think about it, it‘s still a cylinder...that has been curved! Pretty much like a curved tube.

Most shapes that we see on miniatures will be modified basic shapes. However, the light will interact similarly. For example notice, how the thighs in the following pictures respresent cylinders, that are wider at the top. **As the cylinders widen, so do the reflections**.





You‘ll find that many parts of all miniatures represent cylindrical reflection pattern. In the following example, we have goblin‘s arms (especially obvious from the arm guards), squig‘s legs (look at the blue reflection) and even the tongue is formed from 2 connected cylinders.



Most breasts don‘t represent perfect spheres either. You can kinda imagine them as two cylinders connected to half-spheres.

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Reflectivity

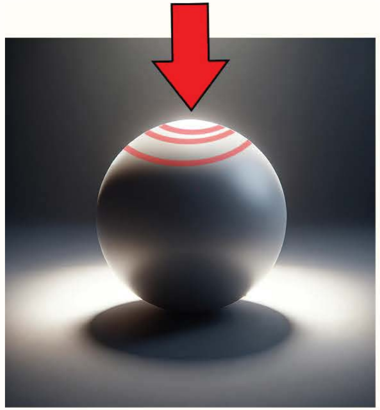
As mentioned on page 3, different materials have different reflectivity and light absorption. Both of the following pictures represent spherical light reflections. We can see that skin reflections are more diffused, the light is less concentrated and there are no secondary reflections. **On less reflective materials, we usually paint with more midtones**.

On the other hand, the armor reflections are more concentrated, there is a prevalance of very light and dark colors with lower amount of midtones. Furthermore, there is a presence of **secondary reflections**. These represent light from the environment. In this case they form half circles at the bottom of the boobplate.

Light direction

For most of my painted miniatures I choose a **„general light“** situation. This assumes light from above the miniature with the light aiming towards the miniature‘s chest and face. Light situation like that makes sure that the most important features are properly exposed. However, if you wanna paint more directional light (or more than one light source), you can do that as well!

We can see how the light situation in both of these images is different, but due to the fact that these are spheres, the light copies circular pattern towards the light source.



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Light direction

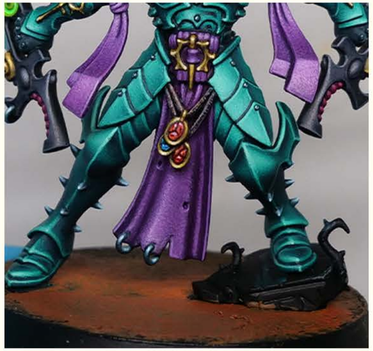
The shapes reflect light in the same pattern no matter the light situation - long shapes (cylinders, cones, etc.) have long reflections, spherical shapes have more round reflections and flat surfaces (cubes, blocks...) usually have a gradient of some kind over the entire surface.

Once we understand this, we can play around with ambient light, directional light and multiple light sources.

In the picture above, the gold armor has painted frontal and side reflections. **The side reflections represent light from the environment and it‘s a frequent feature of metallic surfaces**. Without prior knowledge about reflectivity and shapes, the „Black primer-photo“ method wouldn‘t help us finding them. However, this method does help us find the main reflections, if we assume frontal lighting.

Combining the 2 approaches

Instead of just imagining the light source or just going by the Black primer-photo method, you can **use the best of both worlds**. Take a photo of your miniature in a black primer and if you wanna move the light anyway, go ahead and do that. The underlying knowledge about the reflectivity and how the light interacts with different shapes will help you get there.



In this example I've used both approaches. The fact that I have placed some reflections myself is seen on the kneedpads and the thigh guards (especially the inner ones). The photos with primed miniature don‘t really show any reflections on the knees, aside from the sharp edges.

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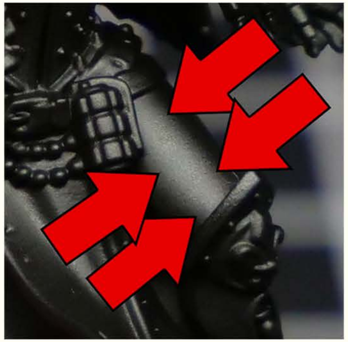
Painting light

To paint the reflections, you start from dark layers and progress with brighter layers towards the reflection position. This is classic layering. With every additional layer you wanna cover less and less so in the end you get a gradient. The more layers you paint between the lightest and the darkest one, the smoother the transition.



Final note

If you wanna get started quickly, there is no problem with just using the black primer photo approach. More complex understanding of shapes will help you with other light directions, not just the frontal one. It's also a good practice to always l**ook at the shapes around you** and see how they reflect the light (quite frequently I stare at the gym equipment because of that).

This tutorial didn't go deep into different materials and their finish, but that is a subject that'll be covered in further guides.

As a refresher, you might wanna watch 2 of my videos on this topic - one about how coloring books help us identify different shapes and the other one about volumetric highlights. This topic is also covered very well in **FAQ2 written by Arnau Lazaro** and published by AK Interactive. You'll find it on page 108.